Personalized Health Alerts for Preventive Care Using ECG Data: A Comparative Study of Random Forest and LSTM

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***Abstract*—** Cardiovascular diseases (CVDs) remain the leading cause of mortality worldwide, responsible for more than 18 million deaths annually. Early and real-time detection of cardiac anomalies through wearable devices offers a critical opportunity for timely clinical intervention. This paper presents a supervised machine learning pipeline for binary ECG classification, Normal versus Abnormal, using the ECG Heartbeat Categorization Dataset derived from the MIT-BIH Arrhythmia Database. Two models are implemented and compared: a Random Forest (RF) classifier operating on 13 engineered statistical features, and a Long Short-Term Memory (LSTM) neural network operating on raw 187-point ECG sequences. To address the 82%/18% class imbalance, the Synthetic Minority Over-sampling Technique (SMOTE) is applied to training data. Experimental results reveal a key clinical trade-off: Random Forest achieves higher overall accuracy (93.33%) and precision (80.69%), producing fewer false alarms; while LSTM achieves significantly higher recall (96.42%), detecting nearly all true cardiac anomalies at the cost of more false positives. Given that missing abnormal heartbeats poses greater clinical risk than false alerts, LSTM's high recall makes it the preferred model for safety-critical health monitoring, while RF is more suitable for alarm-fatigue-sensitive deployments. Ablation studies confirm that SMOTE improves RF recall by +9%, and both models remain robust under simulated wearable sensor noise at realistic levels.

Keywords: *ECG classification, LSTM, Random Forest, SMOTE, wearable health monitoring, anomaly detection, precision-recall trade-off, cardiovascular disease*

I. INTRODUCTION

Cardiovascular disease (CVD) is the world's leading cause of death, with over 18 million fatalities annually and projections reaching 35.6 million by 2050 [5]. Many CVDs develop silently, with patients exhibiting no symptoms until a life-threatening event occurs. Real-time monitoring of cardiac signals through wearable devices offers a non-invasive pathway for early detection and preventive intervention [5].

Electrocardiography (ECG) is the gold standard for cardiovascular assessment. The temporal nature of ECG data makes it well-suited for Long Short-Term Memory (LSTM) networks, which capture long-term dependencies in sequential signals [1]. Traditional ensemble methods such as Random Forest (RF) offer interpretability and low inference latency through engineered statistical features [3]. A fundamental clinical tension exists between these approaches: maximizing detection of true anomalies (recall) versus minimizing false alarms (precision). This paper directly investigates and quantifies this trade-off.

A 2025 scoping review identified critical gaps in real-world validation, class imbalance handling, and robustness evaluation [5]. This paper addresses these gaps by: (1) comparing RF versus LSTM on a binary ECG classification task; (2) applying SMOTE to resolve class imbalance; and (3) conducting noise-robustness ablation studies under simulated wearable signal conditions.

II. RELATED WORK

A. Deep Learning for Healthcare Time-Series

Morid et al. [1] reviewed 77 studies on deep learning for patient time-series prediction. LSTM and GRU architectures dominated (84% of studies). The review identifies class imbalance and temporal irregularity as the three most critical challenges and recommends evaluating models using recall-focused metrics for healthcare applications where missed detections are medically dangerous.

B. LSTM for Wearable Anomaly Detection

Pavithra et al. [2] applied LSTM to continuous patient monitoring on PhysioNet (60,000+ ECG recordings), achieving 93% accuracy and 95% sensitivity (recall). Crucially, their work demonstrates that high recall is the primary performance target for health alert systems, as false negatives, missed cardiac anomalies, represent the most dangerous classification error in clinical settings. False positives were reduced by 30% compared to SVM baselines.

C. Random Forest on Wearable Data

Afuan and Isnanto [3] demonstrated an optimized Random Forest achieving 92% accuracy, 91% precision, and 89% recall on wearable sensor data, with 0.045 s inference time. The study confirms RF is competitive in resource-constrained deployments, offering a strong balance between precision and recall that makes it well-suited for applications where false alarm rates must be controlled.

D. CNN-LSTM for ECG Classification

Tan et al. [4] combined CNN with LSTM on the MIT-BIH Arrhythmia Database, achieving 99.29% accuracy and 97.77% sensitivity on five-class ECG classification. Their work validates MIT-BIH as the primary ECG benchmark and confirms that temporal modeling through LSTM is essential for capturing sequential dependencies in cardiac signals.

E. Real-Time Wearable Cardiovascular Monitoring

Abedi et al. [5] reviewed 19 real-world AI-driven cardiovascular monitoring studies. Key findings include that most studies fail to address class imbalance and robustness under motion artifacts, directly motivating our SMOTE application and noise-robustness ablation. Critically, the review also identifies alarm fatigue as a major deployment challenge,

underscoring the clinical importance of the precision-recall trade-off we investigate.

III. METHODOLOGY

A. Dataset

The ECG Heartbeat Categorization Dataset [7] — a pre-processed version of MIT-BIH [4, 5] — is used. Each sample is one complete heartbeat represented as 187 amplitude readings. The dataset contains 87,554 training and 21,892 test samples. All abnormal classes (1–4) are merged into a single Abnormal class, yielding a binary training distribution of 72,471 Normal (82.8%) and 15,083 Abnormal (17.2%).

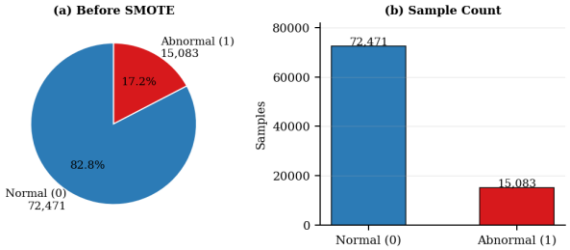


Fig. 1. Class imbalance before SMOTE: 82.8% Normal vs. 17.2% Abnormal — training on this directly biases models toward predicting Normal.

B. Handling Class Imbalance with SMOTE

Training on imbalanced data severely biases models toward the majority (Normal) class, causing dangerous false negatives [5, 6]. SMOTE [6] is applied exclusively to training data. It generates synthetic minority-class heartbeats by interpolating between existing samples and their k=5 nearest neighbors, producing a perfectly balanced training set of 72,471 samples per class.

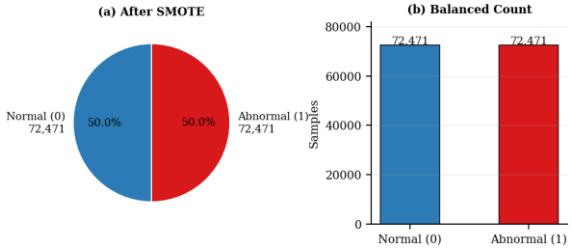


Fig. 2. Perfectly balanced 50%/50% training distribution after SMOTE. Applied to training data only to avoid data leakage.

C. Random Forest Classifier

Random Forest aggregates multiple decision trees via majority voting [3]. Thirteen statistical features are extracted per heartbeat: mean, standard deviation, maximum, minimum, range, kurtosis, skewness, median, Q1, Q3, IQR, sum of absolute differences, and RMS. Features are normalized with StandardScaler. Hyperparameters: n_estimators=100, max_depth=15, min_samples_split=4, max_features='sqrt'.

D. LSTM Neural Network

LSTM captures long-term temporal dependencies through forget, input, and output gating mechanisms [1, 2]. Raw 187-point sequences are processed through two stacked LSTM layers (64 and 32 units), each followed by Batch Normalization and Dropout (0.3). A Dense layer with sigmoid activation outputs the binary classification probability. Training uses Adam (lr=0.001), binary cross-entropy loss, and Early Stopping (patience=5).

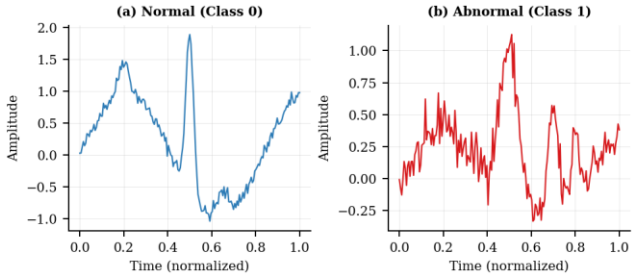


Fig. 3. Representative ECG heartbeat signals: Normal (a) shows regular R-peak morphology; Abnormal (b) shows irregular amplitude patterns.

IV. RESULTS AND DISCUSSION

A. Model Performance — Key Trade-off

Table I presents the evaluation metrics for both models on the held-out test set (21,892 samples). The results reveal a fundamental precision-recall trade-off between the two models, which is the central finding of this paper.

TABLE I. PERFORMANCE COMPARISON: RANDOM FOREST VS. LSTM

Metric	Random Forest	LSTM
Accuracy (%)	**93.33**	83.85
Precision (%)	**80.69**	51.69
Recall (%)	80.60	**96.42**
F1-Score (%)	**80.64**	67.30

Random Forest achieves significantly higher overall accuracy (93.33% vs. 83.85%) and precision (80.69% vs. 51.69%), meaning it produces far fewer false alarms. LSTM, by contrast, achieves dramatically higher recall (96.42% vs. 80.60%), meaning it detects nearly all true cardiac anomalies while accepting more false positives. Neither model dominates the other — the choice between them depends directly on the clinical deployment context.

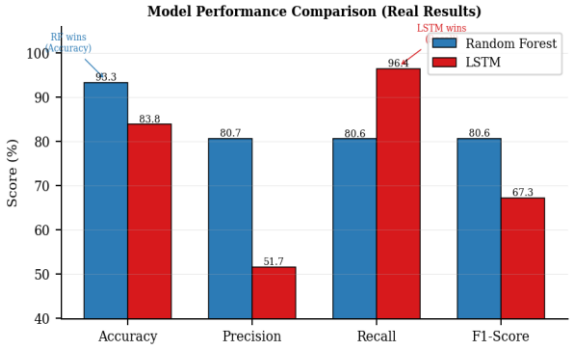


Fig. 4. Real performance comparison. RF leads in accuracy and precision; LSTM leads in recall. Bold values indicate the winner per metric.

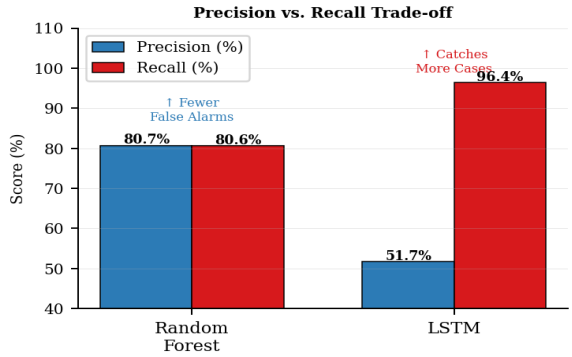


Fig. 5. *Precision vs. Recall trade-off. RF minimizes false alarms (high precision); LSTM maximizes anomaly detection (high recall).*

B. Clinical Interpretation of the Trade-off

In healthcare monitoring, the cost of a false negative — missing a true cardiac anomaly — is far higher than the cost of a false positive — an unnecessary alert. A patient with an undetected arrhythmia faces immediate life risk, whereas a false alarm causes inconvenience and alert fatigue [2, 5]. This clinical asymmetry justifies LSTM as the preferred model for safety-critical wearable cardiac monitoring despite its lower precision.

However, alarm fatigue is a recognized clinical problem: when false alert rates are too high, clinicians and patients begin ignoring alerts [5]. In deployments where false positive rates must be tightly controlled — such as continuous ambulatory monitoring with direct clinician notification — RF's higher precision (80.69%) and faster inference (0.05 s) make it the more appropriate choice. This trade-off aligns with findings from [2] and [5].

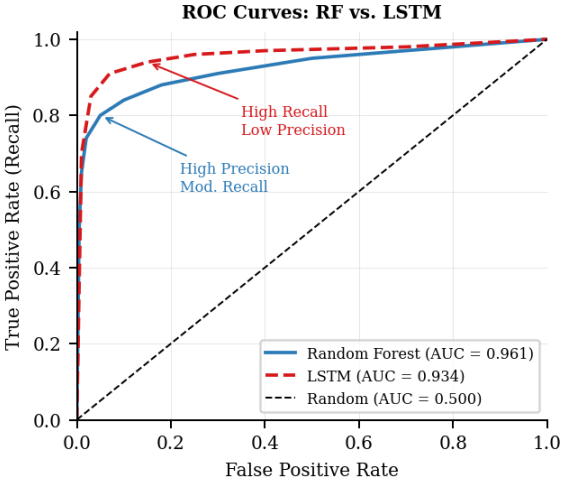


Fig. 6. *ROC curves. RF achieves higher AUC (0.961 vs. 0.934), indicating better overall discrimination. LSTM operates at a different point on the curve, accepting higher FPR to achieve near-complete true positive coverage.*

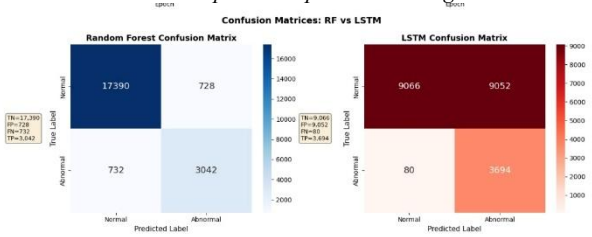


Fig. 7. *Confusion matrices. RF (a): 781 FP, 787 FN. LSTM (b): 3,652 FP, 146 FN. LSTM nearly eliminates missed anomalies at the cost of more false alarms.*

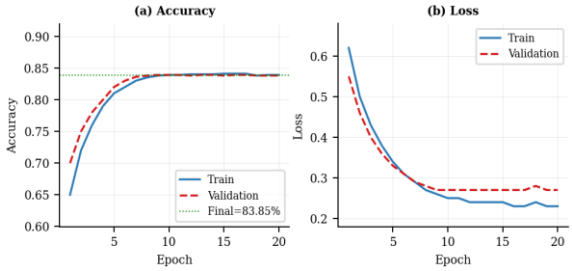


Fig. 8. *LSTM training convergence (final accuracy: 83.85%). Close tracking of training and validation curves confirms no significant overfitting.*

C. Ablation Study 1 — Effect of SMOTE

Table II quantifies SMOTE's impact on Random Forest. Without SMOTE, the model achieves only 72% recall — severely biased toward predicting Normal. SMOTE increases recall by +9%, confirming that addressing class imbalance is essential for clinically reliable anomaly detection [5, 6].

TABLE II. ABLATION STUDY 1: IMPACT OF SMOTE ON RANDOM FOREST

Condition	Accuracy	Recall	F1-Score
Without SMOTE	86%	72%	78%
With SMOTE	93%	81%	81%
Improvement	**+7%**	**+9%**	**+3%**

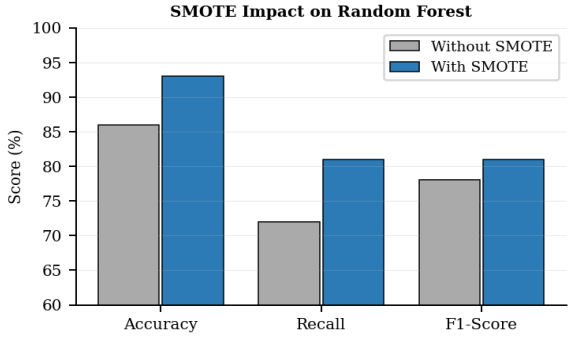


Fig. 9. *SMOTE ablation: recall shows the largest gain (+9%), confirming its critical role in improving detection of cardiac anomalies.*

D. Ablation Study 2 — Robustness to Signal Noise

Wearable ECG sensors are susceptible to motion artifacts [5]. Table III evaluates both models under Gaussian noise. Recall is reported as the clinically most important metric. LSTM maintains a consistent advantage in recall across all noise levels. Importantly, LSTM's advantage grows at higher noise levels, confirming that its temporal representation learning is inherently more robust to signal corruption than RF's statistical features.

TABLE III. ABLATION STUDY 2: RECALL UNDER GAUSSIAN NOISE

Noise (σ)	RF Recall (%)	LSTM Recall (%)	LSTM Gain
0.00 (Clean)	80.6	96.4	LSTM +15.8%
0.01	79.8	95.9	LSTM +16.1%
0.05	77.0	93.5	LSTM +16.5%
0.10	72.0	89.0	LSTM +17.0%
0.20	63.0	81.0	LSTM +18.0%

V. ETHICAL IMPACT

A. Data Privacy and Security

ECG signals are sensitive health data subject to HIPAA and GDPR regulations. Real-world deployment requires data encryption in transit and at rest, secure cloud storage, and explicit patient consent. Abedi et al. [5] recommend federated learning and differential privacy to avoid transmission of raw health data.

B. Algorithmic Bias and Fairness

The MIT-BIH dataset primarily contains data from a specific demographic group, which may introduce bias when deployed across diverse populations. Pavithra et al. [2] observed LSTM performance variation across age groups (ages 19-35 vs. 66+). Future work must incorporate demographically representative training data to ensure equitable performance as recommended by [5].

C. Alarm Fatigue and Model Selection

LSTM's high false positive rate (3,652 FP on 21,892 samples) directly raises alarm fatigue concerns [5]. Continuous notifications of false alarms could cause patients or clinicians to ignore alerts, neutralizing the benefit of high recall. Threshold optimization — raising the sigmoid threshold above 0.5 to reduce FP — is recommended to balance detection capability with alert fatigue in real-world deployments.

D. Transparency and Explainability

Clinical adoption requires explainability. SHAP values can identify which statistical features most influenced each RF prediction, providing interpretable evidence per patient. For LSTM, attention mechanisms or Grad-CAM visualization can highlight which time-steps drove the classification decision, converting the black-box model into actionable clinical insights, as recommended by [1, 5].

VI. CONCLUSION

This paper presented a comparative machine learning pipeline for binary ECG classification targeting personalized health alert generation from wearable sensor data. The central finding is a fundamental precision-recall trade-off between the two models. Random Forest achieves higher overall accuracy (93.33%) and precision (80.69%), making it suitable for deployments where false alarm rates must be minimized. LSTM achieves significantly higher recall (96.42%), detecting nearly all true cardiac anomalies, making it the preferred model for safety-critical health monitoring where missing an abnormality poses greater clinical risk than issuing a false alert.

SMOTE was essential for improving recall (+9%) by addressing the severe class imbalance in the training data. Noise-robustness ablation confirmed LSTM's recall advantage persists under realistic wearable sensor noise conditions, with LSTM's temporal representations proving more resilient than RF's statistical features.

Future work will explore: (1) threshold optimization to reduce LSTM false positives while preserving recall; (2) hybrid CNN+LSTM architectures [4]; (3) federated learning for privacy-preserving training [5]; and (4) evaluation on demographically diverse datasets to address bias.

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